

HAMMADA BALLALI

NATIONALITY: SPANISH – RIGHT TO WORK IN THE UK - ILR

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Experienced IT professional and PhD candidate in Computer Vision with a strong foundation in machine learning and data analysis. Proficient in designing and implementing Python-based solutions, developing ML models, and optimizing workflows for data-driven applications. Skilled in leveraging Python libraries such as TensorFlow, PyTorch, and scikit-learn to build, train, and deploy models for predictive analytics, computer vision, and automation. Experienced in deploying applications on Windows and Ubuntu servers, managing AWS services, and integrating machine learning pipelines into cloud environments. Adept at automating processes using Python scripts and Docker, with hands-on experience in tools like Kafka and Kubernetes. Passionate about AI and machine learning, with a focus on advancing cutting-edge technologies in computer vision to deliver innovative and impactful solutions.

EXPERIENCE

04/2026 – Current

SENIOR INFRASTRUCTURE ENGINEER

CallCare, Manchester, UK

- Designed, built, deployed, and maintained enterprise server and network infrastructure across on-premises and cloud environments.
- Managed and optimized Kubernetes and Linux clusters for high availability, scalability, and performance.
- Deployed and administered containerized workloads using Docker and Kubernetes orchestration.
- Implemented infrastructure monitoring, observability, and alerting solutions using Prometheus, Grafana, and Loki.
- Managed AWS cloud infrastructure and services, supporting secure and scalable production environments.
- Configured and maintained VPNs, tunneling solutions, OpenVPN, Netgate appliances, and Cisco switching infrastructure.
- Installed, configured, and maintained Dell rack servers and enterprise networking hardware.
- Administered Linux and Windows Server environments, including Active Directory, IIS, and Nginx.
- Managed database platforms including MariaDB, MongoDB, and PostgreSQL.
- Supported VoIP and communication systems including Asterisk and 3CX/Web3CX platforms.
- Automated infrastructure provisioning, deployments, and operational tasks using Ansible, Terraform, Bash, and Python scripting.
- Managed identity and access services using Microsoft Entra ID.
- Maintained secure remote access environments using Apache Guacamole.
- Utilized NetBox for infrastructure documentation, IP management, and asset tracking.
- Managed incident response, troubleshooting, and support workflows using Jira and GLPI ticketing systems.
- Collaborated within Agile/Scrum teams to deliver infrastructure improvements and operational enhancements.

- Developed and integrated custom AI-driven tools to improve operational efficiency, monitoring, and automation.
- Performed system performance tuning, patch management, backup strategies, and disaster recovery planning.
- Ensured infrastructure security, reliability, compliance, and operational best practices across all environments.
- Technologies :Kubernetes, Docker, Linux, AWS, Prometheus, Grafana, Loki, Apache Guacamole, MariaDB, MongoDB, PostgreSQL, Microsoft Entra ID, Asterisk, 3CX/Web3CX, Ansible, Terraform, Bash, Python, Jira, GLPI, NetBox, Nginx, Windows Server, Active Directory, IIS, OpenVPN, Netgate, Cisco Switching, Dell Rack Servers, Scrum, AI Automation Tools.

04/2025 – 04/2026

TECHNICAL SPECIALIST L2+ INFRASTRUCTURE ENGINEER

Centred Solutions, Runcorn, UK

- Manage, deploy and update Network and Computing resources.
- Deploying Django applications in Linux Servers
- Administering and managing VMware ESXi and Proxmox hypervisors, for optimal performance, uptime, and security for virtualized environments.
- Manage and monitor and update PostgreSQL servers (Linux – Ubuntu servers)
- Writing and running Ansible scripts.
- Nginx deployment and troubleshooting.
- Manage IT aspects of servers: DNS, routing, interfaces, firewalls and certificates.
- Customize and run PowerShell and Python scripts.
- Ensuring Ensure compliance with internal security policies and regulatory requirements (e.g., GDPR, ISO27001, Cyber Essentials).
- Deploy Docker, Kafka, and Kubernetes.
- Contributed to ClickUp workspace setup and management, optimised workflows, updated user stories, and provided user support and guidance to the development team.
- Support customers in troubleshooting application issues, and root cause analysis as well as monitoring tickets in Odoo.

10/2024 – 11/2025

CODING EXPERTISE FOR AI TRAINING

San Francisco Bay Area (Remote - Freelance)

- Crafting and answering questions related to computer science in order to help train AI models
- Evaluating and ranking code generated by AI models

06/2023 – 04/2025

APPLICATION SUPPORT SPECIALIST – SALES ENGINEERING TEAM

ZOLL Medical, Runcorn, UK

- Deploy and install applications on Windows and Ubuntu servers.
- Handle Active Directory and LDAP communications with application servers.
- Manage and monitor Windows EC2 instances for IIS and SQL servers, as well as Linux EC2 instances in AWS hosting various applications.
- Monitor costs and services in AWS.
- Manage IT aspects of servers, including DNS, routing, interfaces, firewalls, security groups, and certificates.
- Customize and run PowerShell and Python scripts.
- Support Company SDK use for Android, SWIFT, and .NET.
- Deploy Docker, Kafka, and Kubernetes.

- Ensuring Ensure compliance with internal security policies and regulatory requirements (e.g., GDPR, ISO27001, Cyber Essentials).
- Support customers in troubleshooting application servers and SQL servers, as well as monitoring ticketing systems Zendesk/Salesforce/Jira
- Pre-sales support within the Sales Engineering Team

10/2022 -06/2023

TECHNICAL ASSISTANT

Liverpool Hope University, Liverpool, UK

- Assisted primary mathematics department professors in research by processing and presenting data using Tableau, Power BI, and Excel. Created presentations with PowerPoint to effectively communicate findings.

EDUCATION

02/2025 – CURRENT

PHD COMPUTING, UNIVERSITY OF WALES TRINITY SAINT DAVID – (DISTANCE LEARNING BASIS)

- Deep Learning-Based Synthesis of 360° Virtual Reality Photorealistic Scenes from Monocular 2D Videos

10/2024

MASTER OF SCIENCE IN SOFTWARE ENGINEERING AND ARTIFICIAL INTELLIGENCE,
UNIVERSITY OF WALES TRINITY SAINT DAVID

- Dissertation in Deep Learning Model for 12 Lead ECG classification
- Coursework Cloud Computing Docker Compose Deployment of Django/React application in AWS.
- Coursework in AI Meta Ensemble Deep Learning Model for Sentiment Analysis
- Coursework in Software Engineering Agile Plan Conception and Design
- Grade: Distinction

07/2023

BACHELOR OF SCIENCE IN COMPUTER SCIENCE, LIVERPOOL HOPE UNIVERSITY

- Dissertation in Machine Learning: Bellman Ford and Dijkstra Comparison
- Coursework in 3D Unity C#
- Coursework in Machine Learning
- Coursework in MERN stack – Web development
- Coursework in LUA IoT
- Grade : 1st Class A+

PERSONAL PROJECTS

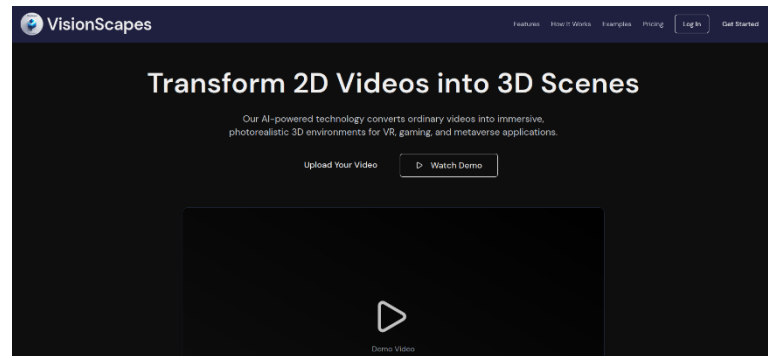
06/2025 – In development

VISIONSCOPES

<https://github.com/HammadaBE/VisionScopes-Demo>

- React · TypeScript · Tailwind CSS · Django · PostgreSQL · Docker · REST API · JWT Authentication
- A full-stack web application designed to enable users to upload monocular video footage of closed spaces and retrieve AI-generated 3D view synthesis through a machine learning pipeline.

- Engineered a React/TypeScript/Tailwind frontend with a Django REST Framework backend, implementing JWT-based user authentication and a media submission system (notes, designed to scale to video uploads)
- Configured a PostgreSQL database with a fully containerised architecture using Docker Compose, enabling one-command local.
- Planned full cloud deployment on Azure/AWS with Nginx as a reverse proxy once the ML model integration is complete
- Designed with scalability in mind: the submission pipeline will feed into a monocular depth estimation and 3D view synthesis ML model.
- Fully testable locally — requires only Docker Desktop; setup and launch instructions are provided in the repository's README.md

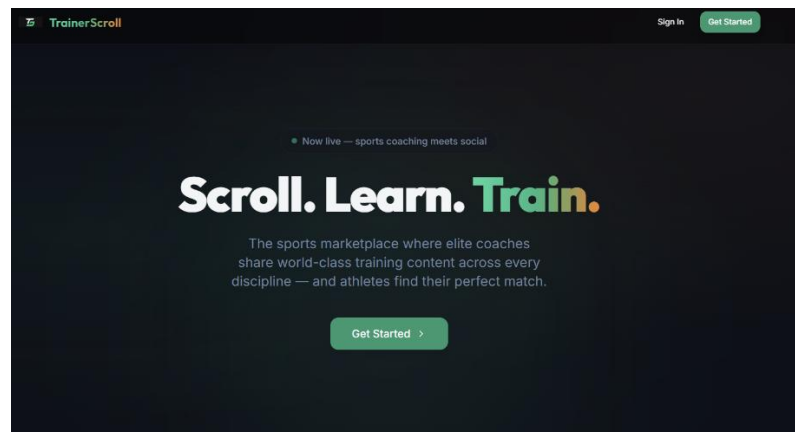


05/2026– In development

TRAINER SCROLL – SPORT ORIENTED SOCIAL NETWORK

<https://trainerscroll.com>

- Next.js, React Native, Supabase, Vercel, JavaScript, TypeScript, REST APIs, Authentication Systems, Cloud Deployment, Mobile Development, Responsive UI Design, CI/CD.
- TailwindCSS



RESEARCH AND PUBLICATIONS

11/2025

- Ebbe, H.B., Vasanwala, M.T.S., Joseph, S. (2025). **An Experimental Comparison of the Effect of Text Preprocessing for Sentiment Analysis Using Ensemble Model**. In: Hewage, C., Kesswani, N., Khan, A., Shekar, B.H. (eds) Responsible Artificial Intelligence. ICRAI 2024. Lecture Notes in Networks and Systems, vol 1504. Springer, Singapore.
https://doi.org/10.1007/978-981-96-8441-0_19

SKILLS

- Google Colab/Jupyter
- Technical documents comprehension
- Customer support needs assessment
- Git/Github
- Salesforce/ Zendesk/Jira
- Python/Keras/TensorFlow/PyTorch
- Power BI/ Tableau/ Excel
- Web Development Django/React
- Databases Windows SQL/Mongo Db
- Could AWS/Azure - Docker/Kubernetes

LANGUAGES

- Spanish: Native
- Arabic: Native
- English: Advanced C1
- French: Intermediate B1

CERTIFICATIONS

- AWS Cloud Practitioner
- JPMorgan Software Engineering VEP

LINKEDIN, GITHUB AND WEB PORTFOLIO

- LinkedIn: <https://www.linkedin.com/in/hammada-ballali-ebbi-4aa5091ba/>
- <https://github.com/HammadaBE>
- Web Portfolio: www.hammadaballali.com